Chapter 6: Conclusions, Recommendations and Future Works

6.1 Introduction

Small language models that run privately on a consumer device know far less than large cloud models, and a frozen model cannot cheaply be retrained to close the difference. This study built CAMR, a memory engine that sits beside a frozen model, and measured what it closes, what it does not, and at what cost, against the alternatives a developer would actually choose between: the model alone, a frontier cloud model, modern retrieval-augmented generation, context stuffing and fine-tuning. This chapter states what the evidence established (section 6.2), what should be done because of it (6.3), and what remains open (6.4). It adds no new results; every statement refers to the section of Chapter 5 that supports it.

6.2 Conclusions

The conclusions answer the research questions of section 1.4 in order, then the questions the later work added: whether answers can be trusted, whether a frozen model can keep learning, and whether automations need such an engine at all.

6.2.1 Research Question i: Which Memory Techniques Work for a Small Model on a Device

Cheap, selective techniques anchored in what the question names work; the weighted composites of large-model agent memories do not. Recency and importance scoring lowered full-support recall on both multi-hop benchmarks, because benchmark questions are independent (section 5.9.2). Entity bridging, a one-step hop that needs no graph, doubled full-support recall on 2WikiMultiHopQA (35.8% → 78.4%) at the same budget, twice what top-10 RAG found with fewer tokens. Gating keeps irrelevant text from a weak reader, and an entity gate stops memory from answering about a similar entity when the right one is missing. Trained retrieval components (a query adapter, a learned index) did not help on unseen questions (section 5.9.8).

6.2.2 Research Question ii: How Much of the Gap a Frozen-Weights Memory Engine Closes

On knowledge, nearly all of it. Eighteen of nineteen small models reached or beat the cloud model on long-tail facts once they had memory (section 5.9.4), and on 200 held-out multi-hop questions a 3B model with CAMR (54.0%) matched the cloud model closed-book (51.8%) at a third of its time (section 5.9.6). On verified answering the 0.5B model was correct 60.5% of the time against the cloud model's 67.5%, while stating a wrong fact 3.0%

of the time against 31.5% (section 5.9.3). Fine-tuning on the same documents reached 13.3% against memory's 71.7% (section 5.9.8).

6.2.3 Research Question iii: Knowledge-Bound Versus Reasoning-Bound Tasks

Memory helps knowledge-bound tasks and not reasoning-bound ones. With the same notes, the gap between a 0.5B and a 20B model was 10.0 points on single facts, 23.3 on multi-hop questions and 83.3 on arithmetic (section 5.9.6), and worked examples lowered the 0.5B model's arithmetic. Where reasoning can be handed to a tool, memory helps again: with a calculator doing the arithmetic, corrections taught the 0.5B model formulas it could not know (0% → 70.8% on invented formulas), and taught the cloud model all ten from four corrections (section 5.9.8).

6.2.4 Research Question iv: The Token Budget

For a 0.5B reader about 128 tokens is best for single facts and 512 for multi-hop questions; more text distracts it (section 5.9.6). The verified pipeline settled at 128 tokens of compressed sentences, and 70% of held-out questions were answered from a fact line with no model call at all. The best budget depends on the reader and the task, which is why the device governor sets it from measured speeds and a latency target (section 5.4.6) and the learned router chooses it per question (section 5.9.12).

6.2.5 Research Question v: Cost on the Device and in the Cloud

On a 4-core CPU the engine needs well under 1 GB beside the model, makes 123,301 paragraphs searchable in under six minutes, and stops cleanly at every device limit (section 5.9.11). Retrieval is cheap in isolation (about 29 ms) but took 56% of a 0.5B model's answer time under contention, so the 10% target was not met for the smallest model. Models of 6B and more answered more slowly than the cloud model on this hardware, which makes 1.5–4B the useful range on a CPU-only laptop (section 5.9.4). In the cloud, the whole study spent $33.36 on 6,456 calls (section 5.9.10).

6.2.6 Beyond the Research Questions

Can its answers be trusted? More than a model's own. Verification cut the 0.5B model's wrong stated answers from 52% to 3.0% on held-out questions, and every remaining wrong answer quoted a stored line; what it cannot catch is a wrong page (section 5.9.3). **Can a frozen model keep learning?** Yes, as long as what it learns is text it reads: from lookups (10.0% → 43.5% on new questions) and from corrections (four corrections taught the cloud model every invented formula, including derived uses no correction showed). Fine-tuning

learned the trained wording better but forgot unrelated skills (section 5.9.8). **Do automations need such an engine?** Where knowledge is private and changes, a model alone is useless or wrong, and any of the three memories made it useful. What separated them was what they did when they did not know: CAMR sent 0.0% wrong replies to customers against 5.0% for RAG and 1.7% for context stuffing, and because it escalated what it did not know, people's answers were taught back and the desk learned; RAG's wrong replies were never seen by a person, so the same wrong answer was repeated (section 5.9.9). With memory, the cloud agent completed 100.0% of its tasks with no hallucinated action; without it, it completed only the tasks that required doing nothing. On news after its training data, the cloud model alone stated a wrong answer to 12.0% of questions; with CAMR's memory of the workflow's pages it was correct on 94.0% with 2.0% wrong.

6.2.7 Contribution and Boundaries

The study contributes: (1) an open-source memory engine whose read path is measured to suit small models; (2) verified answering that states a fact only when memory supports it, with its source; (3) learning without training, by lookup and by correction, measured against fine-tuning and token-level memory; (4) a size map of 22 models (0.5B to 15B) and a reader study showing where memory pays off; (5) a head-to-head comparison with eight modern RAG pipelines on one corpus; (6) automations in n8n and a tool-using agent that show where an engine is needed and what it costs; and (7) CAMR Personal, a released desktop application. The conclusions hold for English questions, frozen quantised models on 4-core CPUs with 16 GB of memory, the sample sizes reported, invented company data and a stochastic cloud reference sampled once. They do not yet extend to real users' data, to phones, or to real customer traffic.

Table 6.1: Alignment from objectives to evidence and conclusions

Objective (section 1.3.2)	**Evidence (Chapter 5)**	**Conclusion**	**Status**
i. Investigate capability gaps and memory techniques	5.9.2, 5.9.6, 5.9.5	Selective, name-anchored techniques transfer; composite scoring does not; the gap is knowledge and reading, not reasoning	Met
ii. Design a lightweight memory engine	5.4, 5.4.4, 5.6.2, 5.6.3	Implemented, tested (203 tests) and changed on evidence; 38 defects found and handled	Met
iii. Develop it on a consumer laptop without a GPU	5.2, 5.9.11, 5.9.13	Runs on 4 cores in under 1 GB beside the model; NFR-02 not met for 0.5B	Met on laptop-class machines, not one physical laptop
iv. Evaluate gap closed by task type	5.9.4, 5.9.6, 5.9.3	Facts closed; multi-hop matched from 3B; reasoning not closed	Met
v. Determine the best	5.9.6, 5.9.12	128 tokens (facts), 512 (multi-hop) for	Met

Objective (section 1.3.2)	Evidence (Chapter 5)	Conclusion	Status
token budget		0.5B; set per device and per question	

6.3 Recommendations

The first table is a decision guide: which approach to use, given the situation, on the evidence of this study. The second lists recommendations by audience.

Table 6.2: Decision guide: memory engine, RAG, context stuffing, fine-tuning or a cloud model

Situation	Use	Why (Chapter 5)
Private, changing knowledge (support, operations); a wrong answer has a cost	CAMR with the strongest reader allowed; escalate what is not supported and teach the answer back	5.9.9: 0 wrong replies; the desk learned from its escalations
A small, stable knowledge base and a cloud model	Context stuffing is simplest, but costs more per question as the base grows and still states wrong answers	5.9.9, 5.9.10
A large document collection; coverage matters more than an occasional wrong answer	Hybrid search with a reranker (modern RAG)	5.9.5
The same, but wrong answers are costly	Modern RAG retrieval with CAMR's check, or CAMR	5.9.5
Questions that chain two entities	CAMR's entity bridging with a reader of 3B or more	5.9.2, 5.9.6
Arithmetic or multi-step reasoning	A stronger model, or a calculator tool; not memory	5.9.6, 5.9.8
A new rule or formula learned from a few corrections	Store the rule and a worked example; let a calculator do the arithmetic	5.9.8
A fixed output format, and nothing else must be kept	Fine-tuning (LoRA) can win on that format	5.9.8: 88.3% vs 50.0%, but GSM8K 13.3% → 0%
Offline or private use on a 16 GB CPU laptop	1.5B model for facts, 3B for reading across documents, with CAMR; not 6B or larger	5.9.4
Events after the model's training data	Teach the source as it arrives (an automation can do it) and answer from memory	5.9.9

Table 6.3: Recommendations

#	Audience	Recommended action	Evidence and condition
1	Developers of on-device assistants	Pair a 1.5–4B model with external memory for personal and long-tail facts; do not rely on the model's own knowledge	5.9.4: alone ≤ 33%, with memory 70–80%; faster than the cloud on 4 cores
2	Anyone who lets a model answer customers	State only what memory supports, show the source, and route the rest to a person; measure wrong replies, not only accuracy	5.9.3, 5.9.9
3	Builders of automations (n8n, agents)	Put company knowledge behind a memory with an escalation path, and teach every human resolution back into it	5.9.9: the learning loop only works when unknowns are escalated
4	Developers	Send arithmetic and multi-step reasoning to	5.9.6: 83-point residual; 5.9.8: calculator

#	Audience	Recommended action	Evidence and condition
		a stronger model or a tool, not to memory	
5	Designers of memory engines	Rank by similarity with gating, entity bridging and an entity gate; avoid recency weighting for independent questions; tune budgets to the reader	5.9.2, 5.9.12
6	Teams choosing between fine-tuning and memory	Use memory for knowledge that changes or must be deleted; fine-tune only for a fixed output format where nothing else must be kept	5.9.8: LoRA lost GSM8K (13.3% → 0%)
7	Researchers	Report hallucination (wrong stated answers) beside accuracy, judge aliases, and use held-out sets answered once	5.9.3, 5.9.18
8	The student and supervisor	Obtain ethics approval before collecting feedback from real users of CAMR Personal; rotate the cloud key used in this study	5.6.4, 5.7

6.4 Future Works

Each direction follows from a failure or an open question in section 5.9.17 or 5.9.18, ordered by expected value.

Table 6.4: Prioritised future works

Priority	Direction and rationale	How to evaluate
1	**A user study of CAMR Personal.** Whether the assistant grows more useful as a person's memory grows is its central claim, and no user has tested it.	4–8 weeks, ethics approval; share of answers from memory, thumbs-up rate, accuracy on each user's repeated questions
2	**A real support queue.** The desk used an invented company; a real queue (with consent and anonymisation) would test coverage, escalation load and wrong replies at scale.	Wrong-reply rate and escalations per 1,000 tickets against the current process
3	**Better checks.** The kind check withheld correct answers (*al-Ghazali*), and the check cannot tell a wrong page from a right one.	Coverage and wrong-stated rate on the held-out set; source agreement across two pages before stating
4	**A reranker in CAMR's recall, and a stricter claim check.** Reranked RAG put the answer in front of the reader more often (72% against 64%), and the 0.5B reader got answers past the check that only repeated the subject's name.	Correct and wrong-stated rates on the held-out set and the desk
5	**Better lookups.** The answer was in what was read for 52–73% of questions: the lookup sets the ceiling.	Answer-in-text rate on a fresh stream; Wikidata for disambiguation
6	**Closing the reading gap of small models.** 76% of the 0.5B model's multi-hop misses had the answer in context.	Correct-when-in-context on HotpotQA and 2Wiki, $n \geq 200$
7	**A learned entity recogniser.** The rule-based entity gate caused D-15, D-31, D-32 and D-37.	Abstention on answerable and unanswerable questions; full-support recall
8	**Sessioned memory.** Recency could not help on independent questions; conversations are not independent.	A sessioned benchmark (LongMemEval or LoCoMo) with and without recency
9	**Phones and a physical laptop.** Timings came from laptop-class cloud machines.	NFR-02, peak memory and battery per 100 answers on two phones and one laptop